

Module 9: Outlier Detection

- ❑ Basic concepts
- ❑ Statistical approaches
- ❑ Proximity-based approaches
 - ❑ Distance-based outlier detection
 - ❑ Density-based outlier detection
- ❑ Reconstruction-based approaches
- ❑ Clustering and classification based approaches
- ❑ Mining contextual and collective outliers
- ❑ Outlier detection in high-dimensional data

Proximity-Based Approaches

- ❑ Intuition: objects that are far from others can be regarded as outliers
- ❑ Two types of proximity-based outlier detection methods
 - ❑ Distance-based methods: an object is then considered as an outlier if its neighborhood does not have enough other points
 - ❑ and density-based methods: an object is identified as an outlier if its density is relatively much lower than that of its neighbors